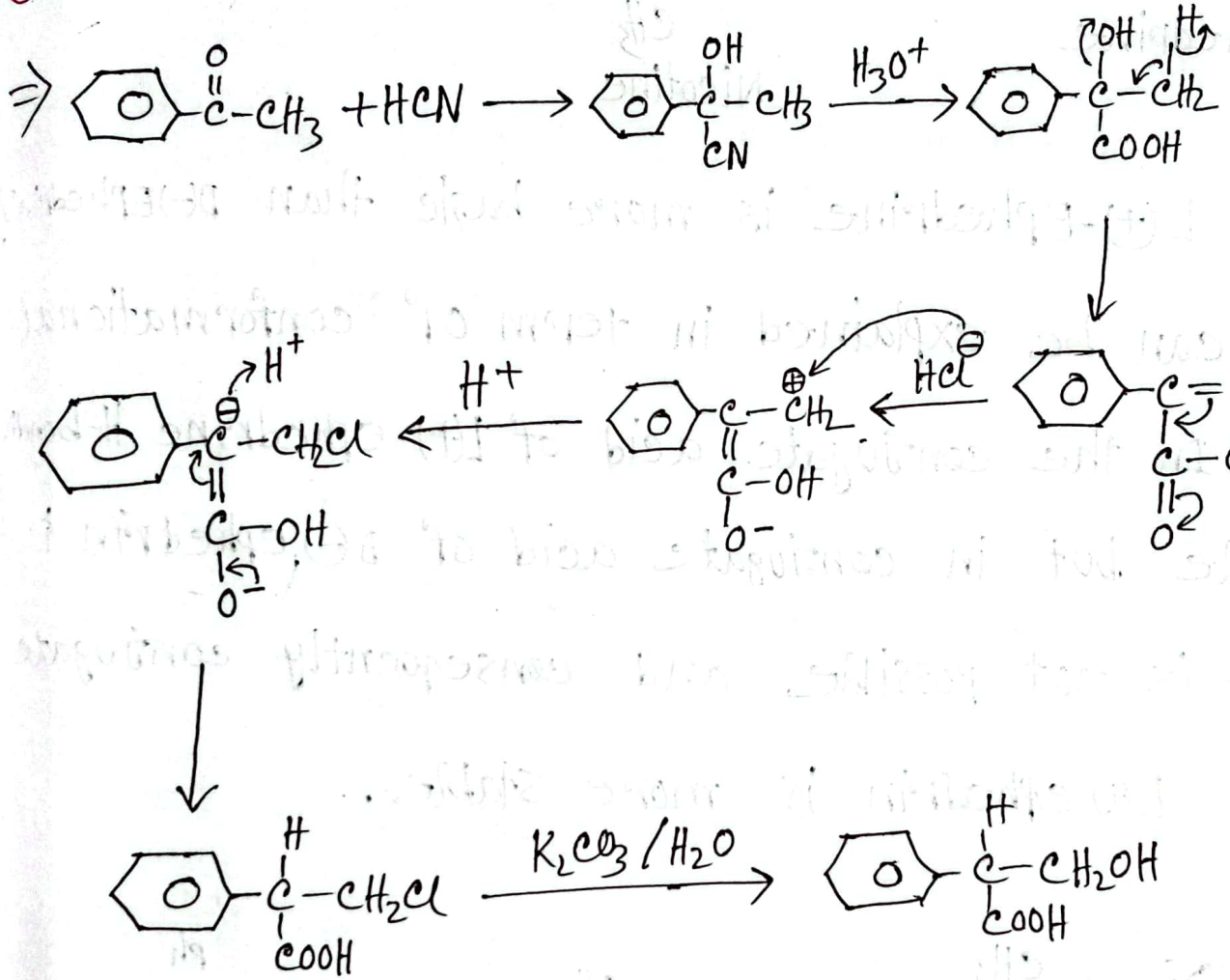
Alkaloid

Rajib
Che-50

(*) 2018, 2022(t) Outline the synthesis of tropic acid?

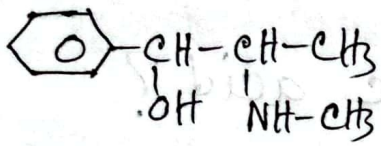


Tropic acid

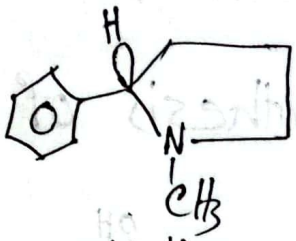
(*) 2022(t) Write the general properties of alkaloid?

⇒ ① colourless ② crystalline ③ Non-volatile ④ optically active ⑤ Have bitter taste ⑥ Insoluble in H_2O but soluble in other organic solvents. ⑦ They are generally tertiary nitrogen in ring system. ⑧ Most of the

alkaloids contain oxygen. EX.



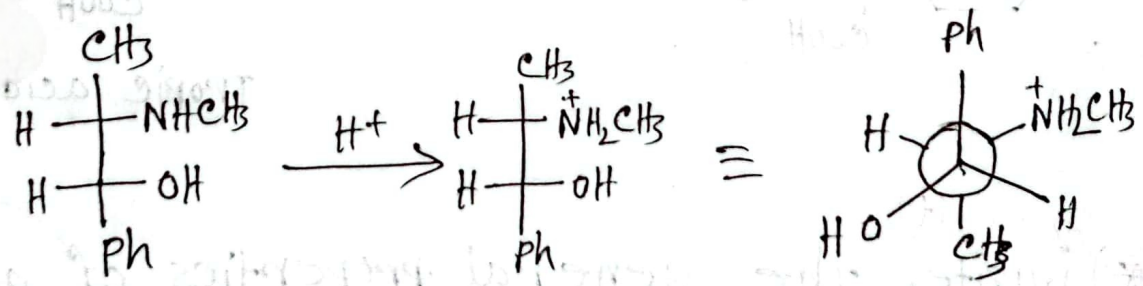
Ephedrine



Nicotine

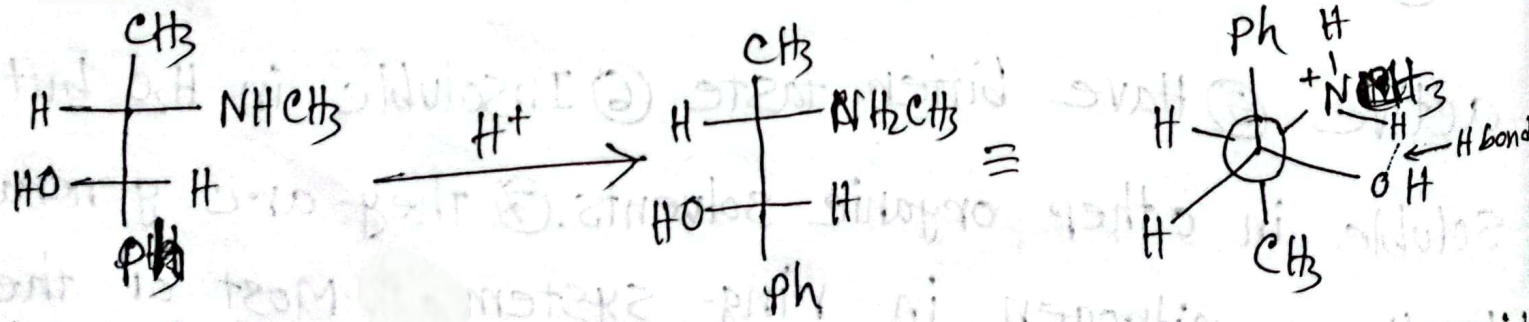
2021 why L(+)-Ephedrine is more basic than D(-)Ephedrin?

⇒ This can be explained in term of conformational analysis. In the conjugate acid of L(+)-ephedrine H-bondin is possible but in conjugate acid of D(-)ephedrin H-bonding is not possible and consequently conjugate acid of L(+)ephedrin is more stable.



D(-)-Ephedrin

unstable conjugate aci



L(+)ephedrin

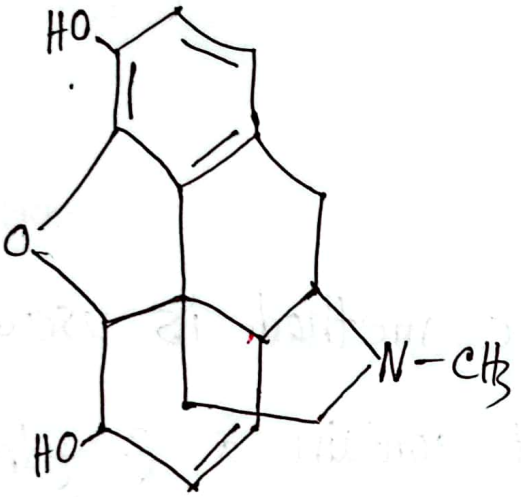
Stable conjugate acid

AS L(+) ephedrine form more stable conjugate acid than D(-) ephedrine so L(+) ephedrine ($pK_a = 9.22$) is more basic than D(-) ephedrine ($pK_a = 9.19$).

2021, 2016

⋆ Draw the structure of morphine?

⇒



morphine

2022, 2022(t)

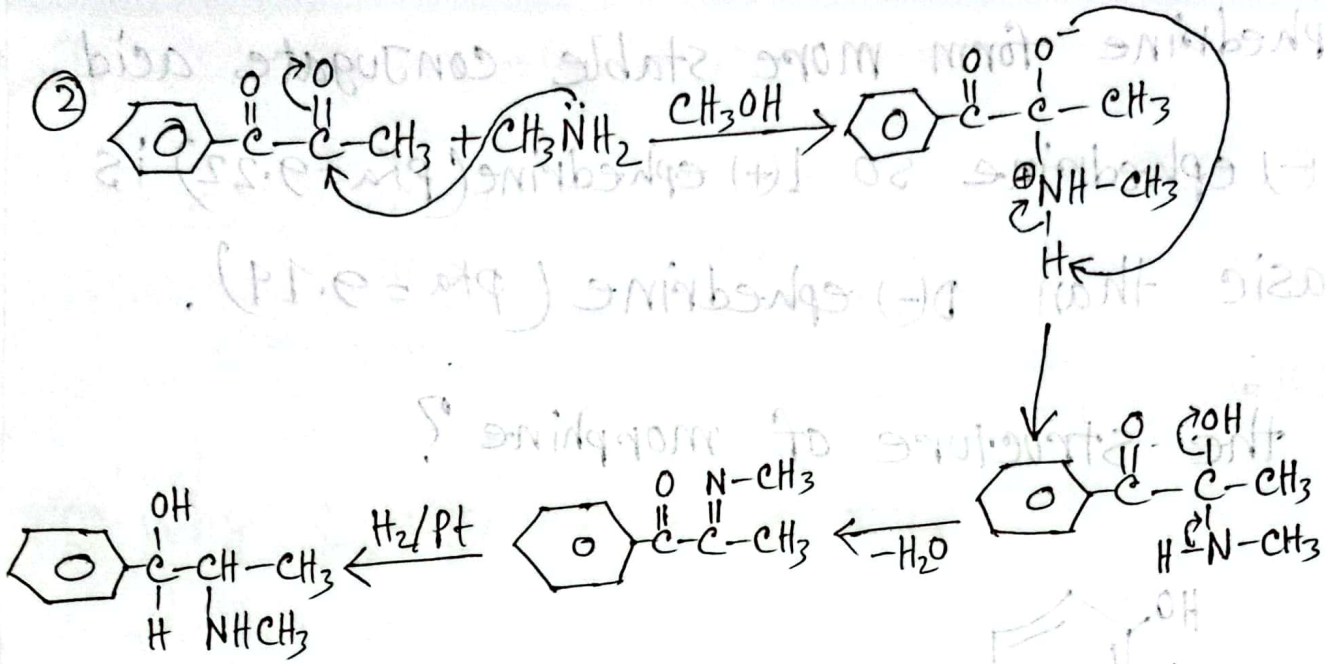
⋆ synthesis Ephedrine using two different methods?

⇒ (I) $C_6H_5CHO + C_2H_5NO_2 \xrightarrow{K_2CO_3} C_6H_5-\underset{OH}{CH}-\underset{NO_2}{CH}-CH_3$

$\downarrow H_2/Pt$

$C_6H_5-\underset{OH}{CH}-\underset{NH_2}{CH}-CH_3 \xrightarrow{CH_3I} C_6H_5-\underset{OH}{CH}-\underset{NHCH_3}{CH}-CH_3$

Ephedrine ($C_{10}H_{15}NO$)



Ephedrine ($C_{10}H_{15}NO$)

2021

(X) When Emde's degrative method is used & why?

⇒ If the alkaloid doesn't contain a β-hydrogen atom, the hoffman's exhaustive methylation method fails, In such case Emde's method may be employed.

⇒ this method is used to degrade the ring- structure of these compounds to achieve a more straightforward or smaller molecule, which can then be further studied or used in synthesis. The emde degradation method is relatively specific for demethylation of tertiary amines. The emde degradation can proride good yields of the desired secondary amine product. For these reason these method

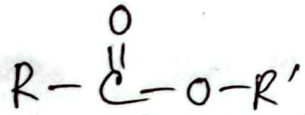
is used.

2021
Prove that Atropine is an ester?

⇒ ① Atropine is a tropane alkaloid that consist of two main component: (i) Tropic acid: A carboxylic acid derivative.

(ii) Tropine: An alcohol derivative.

In atropine, the hydroxyl (-OH) group of tropine forming an ester bond with the carboxyl group (-coo) of tropic acid. This ester linkage classifies atropine as an ester. structure of atropine:



② Hydrolysis: Atropine undergoes hydrolysis in the presence of a strong base or acid.

$$C_{17}H_{23}NO_3 + H_2O \xrightarrow{Ba(OH)_2} \underset{\text{Tropic acid (-COOH)}}{C_9H_{10}O_3} + \underset{\text{Tropine (-OH)}}{C_8H_{15}NO}$$

The formation of these two products is a clear indicatio that atropine contains an ester bond.

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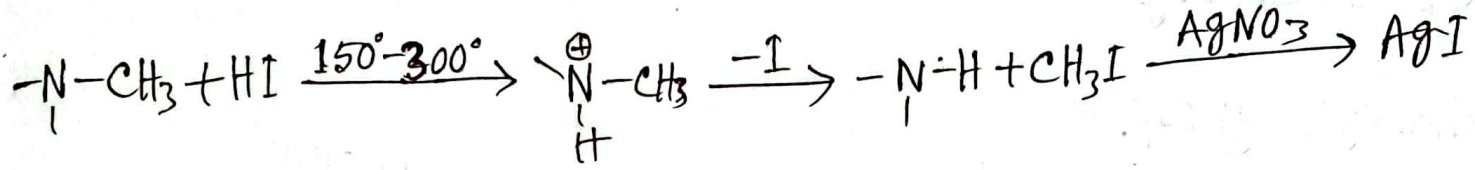
③ Atropine exibits similar properties of ester such as sweet, fruity odors, relatively low b.p., solubility in organic solvent but not in water. These property supporting Atropine's ester nature.

④ The IR spectrum of atropine shows a characteristic peak at around 1730 cm^{-1}, which is attributed to the carbonyl (>c=o) stretching vibration of the ester group. AS carbonyl stretch (>c=o) in esters appears around 1735-1750 cm^{-1}.

⑧ 2021 Explain Herzig-Meyer method for determinig N-CH_3 groups in alkaloid.?

⇒ The Herzig-Meyer method is based on the conversion of the N-methyl group into CH_3I using HI. Firsly the alkaloid is treated with excess HI. This reaction forms CH_3I. Then CH_3I, being volatile, is distilled from the reaction mixture. The distilled CH_3I passed

into a sln of $AgNO_3$ & In this reaction AgI is formed the amount of AgI formed is directly proportional to the amount of CH_3I, which correspond to the number of N-methyl groups in the original alkaloid.



The molar relationship btwn CH_3I and AgI is 1:1. So from the mass of AgI the number of $N-CH_3$ groups in the sample can be calculated.

2021
Elucidate the structure of Ephedrine?

⇒ 1. Molecular formula: on the basis of analytical data & molecular weight determination, the molecular formula of ephedrin is $C_{10}H_{15}NO$.

2. Presence of secondary amino group: Ephedrin was treated with Nitrous acid gives nitroso compound which shows the presence of secondary amino group.

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$$C_{10}H_{14}ONH + HONO \longrightarrow C_9H_{11}O\underset{\underset{NO}{|}}{N}-CH_3 + H_2O$$

3. presence of -OH group: on benzoylation ephedrine gives dibenzoyl derivative. This show the presence of hydroxyl group.

$$C_{10}H_{13}\begin{bmatrix} OH \\ NH \end{bmatrix} \xrightarrow{2C_6H_5COCl} C_{10}H_{13}\begin{bmatrix} OCOC_6H_5 \\ NCOC_6H_5 \end{bmatrix}$$

ephedrine — Dibenzoyl derivative of ephedrine.

→ 2016, 2015, TU

4. presence of N-methyl amine: on boiling with conc. HCl ephedrine gives methyl amine & ethyl phenyl Ketone showing the presence of $N-CH_3$ group.

$$C_{10}H_{15}NO \xrightarrow[\text{conc. HCl}]{\Delta} CH_3-NH_2 + C_6H_5COCH_2CH_3$$

methyl amine

→ 2016 (t)

5. presence of benzene ring: Ephedrine upon oxidation gives benzoic acid indicates that it contain monsubstituted benzene ring with a side chain.

$$C_{10}H_{15}NO \xrightarrow[\text{oxidation}]{KMnO_4} \text{C}_6\text{H}_5\text{COOH}$$

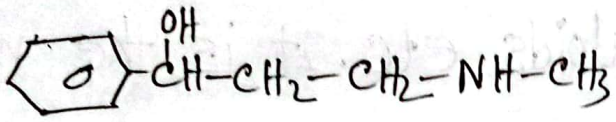
Benzoic acid

~~6. Nature of side chain:~~

6. Nature of -OH group: Ephedrine react with CrO_3 and form Ketone. This indicate presence of 2° Alcohol.

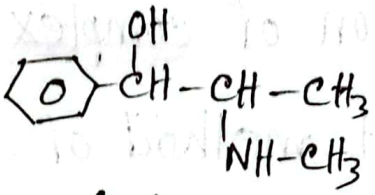
$$C_{10}H_{15}NO \xrightarrow{CrO_3} \text{Ketone}$$

So possible structure:



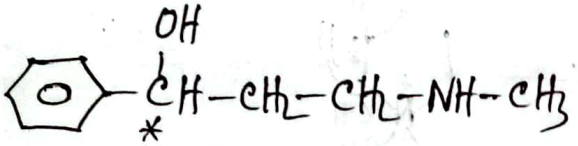
(I)

or



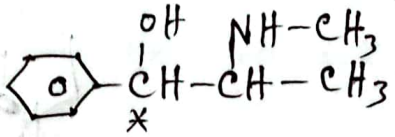
(II)

we know Ephedrine is optically active. In Real ephedrine sample if replaces -OH of chiral center by -H then also it will be optically active.



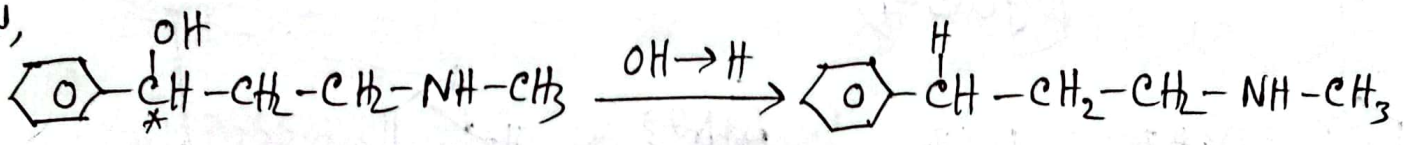
(I)

or



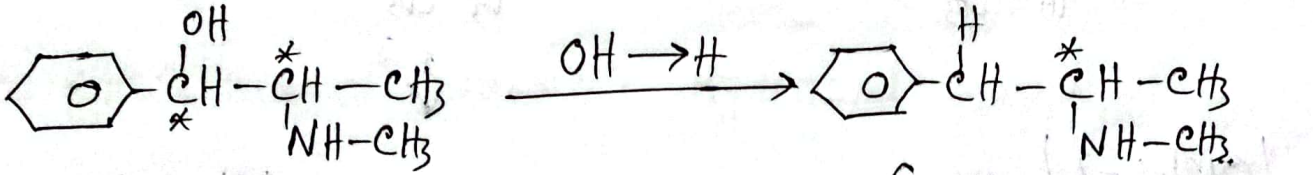
(II)

Now,



(I)

(optically inactive)



(II)

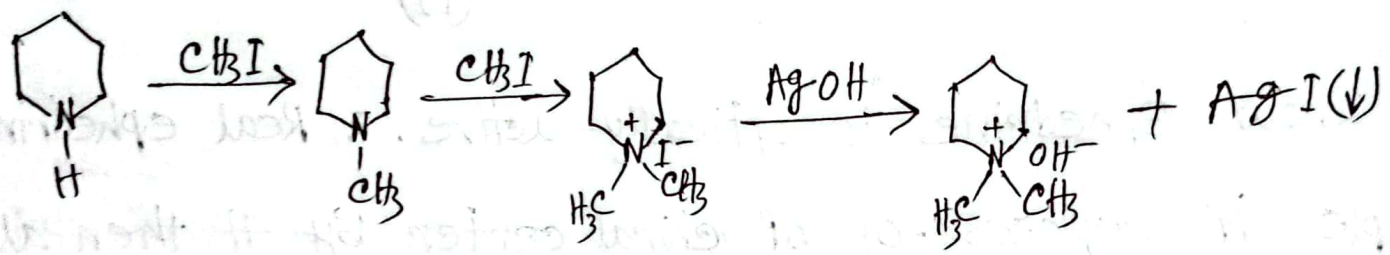
(optically active)

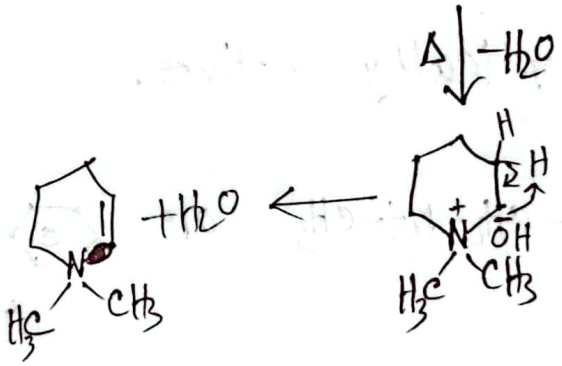
So the real structure will be II.

2020, 2016(t)

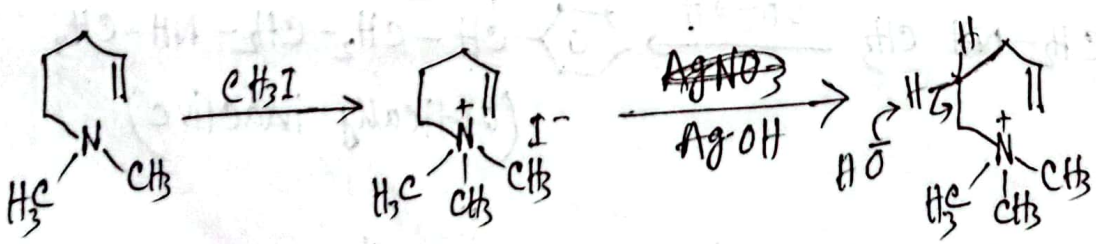
What are the importance of Hofmann's exhaustive method with example?

⇒ Importance are: Structural Elucidation, Functional group determination/identification, Enhance solubility and stability, synthesis of quaternary ammonium compounds degradation of complex alkaloids etc. It is the most important method of opening heterocyclic rings. example

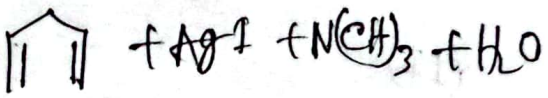




(cycle-1)



(cycle-2)



2018, TU

(*) What product would you expect to be formed when ephedrine is treated with conc. HCl?

⇒ when ephedrine react with conc. HCl it gives methyl amine & ethyl phenyl Ketone showing the presence of N-CH_3 group.

$$C_{10}H_{15}NO \xrightarrow[\Delta]{\text{conc. HCl}} CH_3-NH_2 + C_6H_5COCH_2CH_3$$

2018, 2017, 2016, 2015, 2022, TU

(*) Morphine contains two -OH & one of the -OH is phenolic in nature - explain?

⇒ Morphine reacts with halogen acid gives monohalogen derivative. In this derivative one hydroxyl group of C-6 is replaced by halogen acid this reaction is characteristic of alcohol.

Morphine + HCl → Morphine hydrochloride (a salt)

An when Dissolve morphine in water and add $FeCl_3$ then we found violet or green colored complex which confirms the presence of phenolic hydroxyl group.

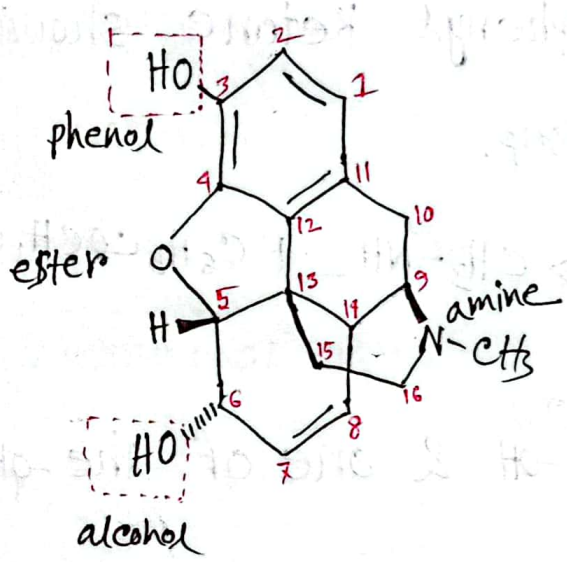
Also, Morphine react with acetic anhydride and forms a diacetyl derivative which indicates the presence of

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two hydroxyl group.

Morphine + $(CH_3CO)_2O$ ⟶ Diacetylmorphine + acetic acid

structure of morphine :



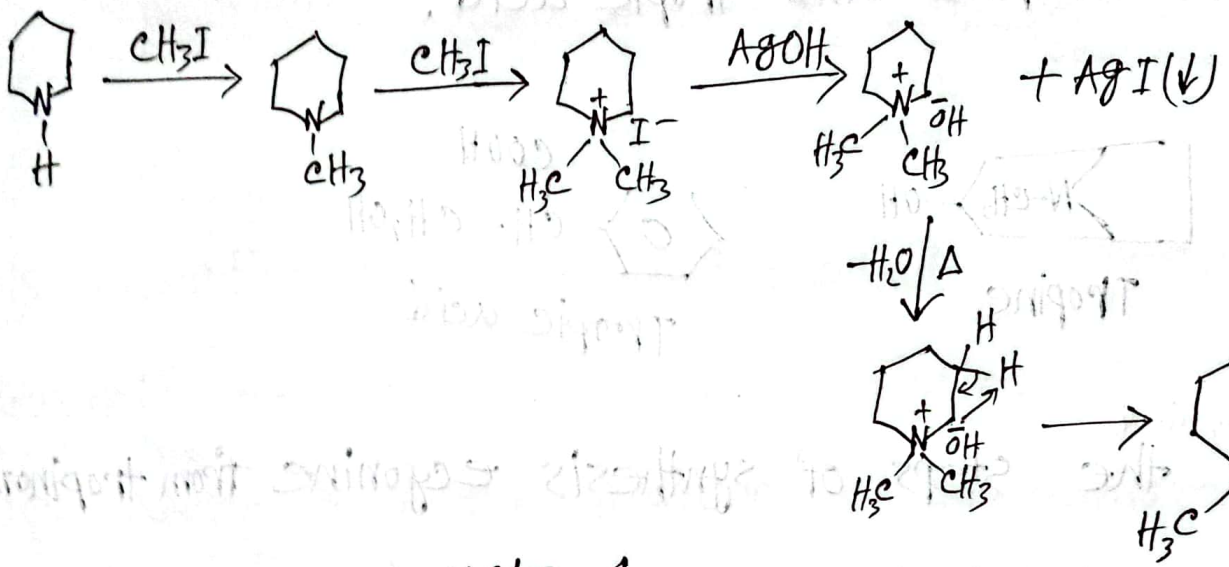
2018, TU

(X) Discuss Hofmann's exhaustive methylation method ?

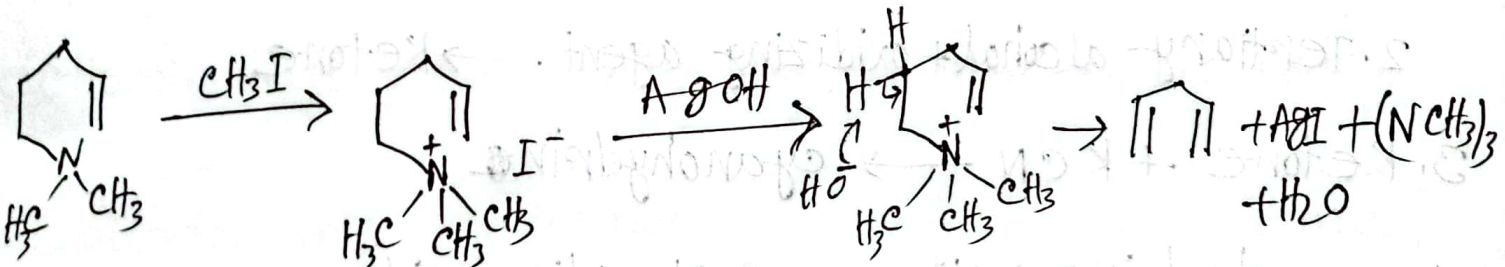
⇒ Hofmann's e.m. is a chemical reaction used to degrade amines to produce alkenes. This method is particularly usefull for determining the structure of amines.

Exhaustive methylation is the process whereby amine reacts with excess methyl iodide to produce quarternary ammonium salt, an unstable intermediate molecule with a positive charge. Then the hydroxyl ion takes the place of the iodide ion. AgI is created when the iodide ion interacts with the silver ion. The creation

of an alkene and tertiary amine is the final step in hofmann's reaction.



cycle-1



cycle-2

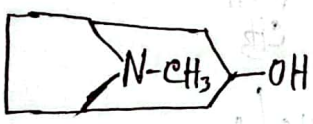
2017, 2019(t)

✱ what product formed when atropine is warmed with aq. $Ba(OH)_2$ solution? Draw the structure of product?

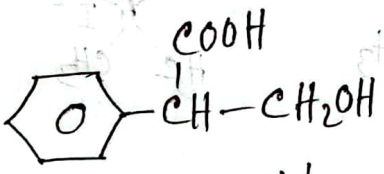
✱ Atropine is an ester of tropic acid and tropine so when Atropine treated with a strong base like barium Hydroxide ($Ba(OH)_2$), the ester bond is cleaved and produces tropic acid and tropine.

$$C_{17}H_{23}NO_3 + H_2O \xrightarrow{Ba(OH)_2} \underset{(C_9H_{10}O_3)}{\text{Tropic acid}} + \underset{(C_8H_{15}NO)}{\text{Tropine}}$$

⇒ Structure of tropine and tropic acid:



Tropine



Tropic acid

⊛ 2017 (mam porainai) outline the steps of synthesis ecgonine from tropinone

⇒ 1. Tropinone + CH_3MgBr ⟶ Tertiary alcohol

2. Tertiary alcohol + oxidizing agent ⟶ Ketone

3. Ketone + KCN ⟶ cyanohydrin

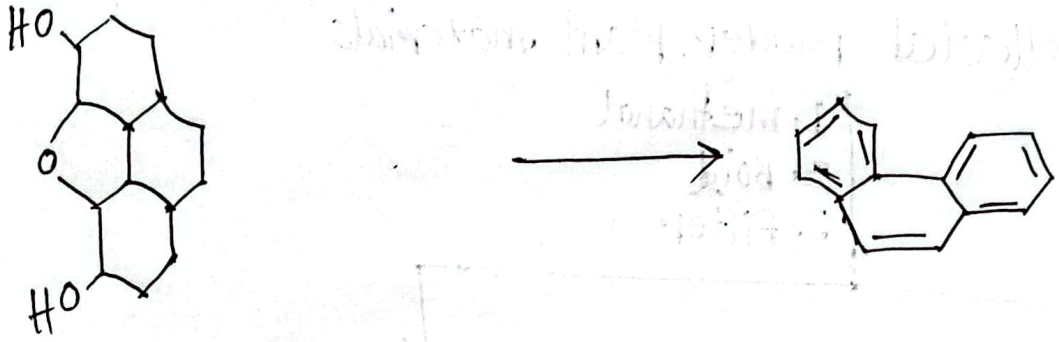
4. cyanohydrin + H^+ ⟶ carboxylic acid

5. carboxylic acid + CH_3OH + H_2SO_4 ⟶ Ecgonine

⊛ 2017, 2016, 2015 Prove there is a phenanthrene nucleus in the morphine molecule?

⇒ When morphine is distilled with zinc dust, it yields phenanthrene and a number of bases, suggesting that morphine may contain phenanthrene nucleus.

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this reaction indicates that morphine is possessing phenanthrene nucleus. Also by analysing the structure we can identify the presence of phenanthrene skeleton. Morphine undergoes demethylation & rearrangement reactions that also reveal the phenanthrene core.

(8) 2015 Indicate the medicinal uses of D(-) ephedrine?

⇒ D(-)-Ephedrine, also known simply as ephedrine, is a naturally occuring alkaloid which is particularly uses in the respiratory and cardiovascular system.

(i) Ephedrine is used to treat low blood pressure, especially during anesthesia.

(ii) Ephedrine is a mild central nervous system stimulant.

(iii) It is commonly used as a bronchodilator to treat conditions like asthma & bronchitis.

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(A) 2022, 2015, 2020, 2018, 2019
Discuss the extraction & isolation of alkaloids ?

⇒ collected powder, plant materials
1. methanol
2. Boil
3. Filter

Filtrate methanol ← extraction of methanol — Solid mass
(alkaloids/methanol soluble organic compound + non alkaloid Basic compound + other neutral compound + acidic organic compound)

5% HCl + 95% H_2O

Salt of basic compounds (alkaloid/non alkaloid) + water soluble org compound + Solid mass

Filter

Solid ~~layer~~ mass + Aq. layer
(salt of basic org compound)

$CHCl_3$

Aq. layer
(salt of basic compound)

$NaHCO_3$ / Na_2CO_3

Basic compound (alkaloid + non alkaloid + inorganic salt)

$CHCl_3$

$CHCl_3$ layer

Aq. layer

$CHCl_3$ layer
(water soluble org compound)

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$CHCl_3$ layer

(free alkaloid + non alkaloid base)

↓ ① Anhydrous Na_2SO_4
↓ ② Evaporation

Solid/semisolid mass

↓ chromatographic separation (column)

individual free alkaloid.

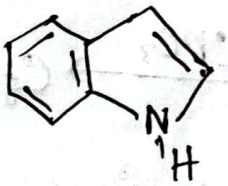
2015
(*) What are the main characteristic of plant alkaloids?

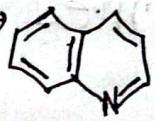
⇒ ① crystalline, ② colorless ③ Non volatile ④ Insoluble in water but soluble in other solvents ⑤ Have bitter taste ⑥ optically active ⑦ Most of the alkaloids contain oxygen. ⑧ Alkaloids are generally basic compounds due to the presence of amine groups.

2022 (t)
(*) classify alkaloid A. to the nucleus present in the molecule

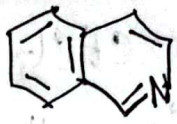
⇒ classification of alkaloid:

1. Phenylethylamine alkaloids → phenylethylamine group → 

2. Indole alkaloids → Indole ring → 

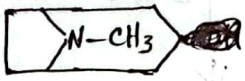
3. Quinoline alkaloids → Quinoline ring → 

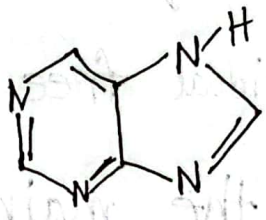
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4. Isoquinoline alkaloid → Isoquinoline ring → 

5. pyridine alkaloid → pyridine ring → 

6. piperidine alkaloid → piperidine ring → 

7. Tropane alkaloid → Tropane ring → 

8. purine alkaloid → purine ring → 

9. pyrrolidine alkaloid → pyrrolidine ring → 

2022(t), 2022

Elucidate the structure of tropine?

⇒ 1. Molecular formula: $C_8H_{15}NO$

2. DBEV = 2, that means no benzene ring in this structure

3. $C_8H_{15}NO \xrightarrow[CCl_4]{Br_2}$ No reaction

this reaction prove, No double/triple bond in this compound

4. $C_8H_{15}NO \xrightarrow{CrO_3} C_8H_{13}NO$ (Ketone)

As Ketone is formed so 2° Alc is present.

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5. $C_8H_{15}NO \xrightarrow{Ac_2O}$ Mono-acetylated derivative

As Mono-acetylated derivative is produced, it prove that "N" is in tertiary form.

6. $C_8H_{13}NO \xrightarrow{[O]} C_8H_{13}NO_4$

Tropinone — Tropic acid

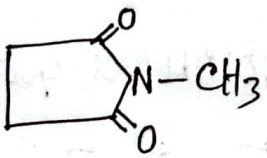
carbon no. is same on both side, so the 2° OH is inside the ring.

7. $C_8H_{15}NO \xrightarrow[<150°C]{HI} C_8H_{14}NI \xrightarrow[HCl]{Zn} C_8H_{15}N \xrightarrow{HCl} C_7H_{13}N + CH_3Cl$

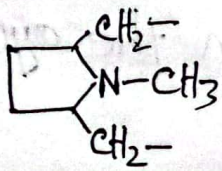
Tropine

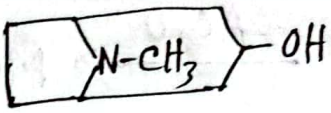
Since one CH_3Cl is produced, N is attached with one $-CH_3$ group.

8. when two moles of benzaldehyde is added to tropine, dibenzilidine derivative is produced. So, two methylene group is present in the vicinity of 2° $-OH$ group.

9. $C_8H_{13}NO \xrightarrow[H_2SO_4]{CrO_3}$ 

(N-methyl succinamide)

So,  is present in tropine structure.

So, the final structure is 

2018, TU
What are alkaloids?

⇒ The nitrogeneous heterocyclic organic compound found majorly in plants & animal product, usually having basic nitrogen containing complex structure, physiologically active are generally called alkaloids.

2022
How can you identify 3° amine in cyclic form of alkaloid?

⇒ ① 3° amine do not react with nitrous acid (HNO_2) But 1° and 2° react so we can identify 3° amine by this reaction

$$3° \text{ Amine} + HNO_2 \longrightarrow \text{No reaction}$$

② 3° amine cannot be alkylated because the lack of hydrogen atom on the Nitrogen. But 1° & 2° undergo alkylation.

③ If the alkaloid contains a tertiary amine, it will react with the alkyl halide to form a quarternary ammonium salt.

$$R_3N + CH_3I \longrightarrow [R_3N^+CH_3]I^-$$

Here, R_3N represents the tertiary amine in the cyclic alkaloid.

CamScanner

(★) what is the source of ephedrine? Briefly describe one suitable method for extraction of ephedrine?

⇒ ephedrine is a naturally occuring alkaloid. in several plant species such as ephedra sinica, ephedra nevadensis, Ephedra equisetina, ephedra viridis, ephedra intermedia etc.

⇒ Take weighed quantity of ~~young~~ dried young stems of ephedra & powder to obtain coarse sized particles. Extract with petroleum ether. Filter & discard ether extract. Dry the marc & moisten with dilute NH_3. Extract this marc with aq. alcohol. Filter & evaporate Filtrate to obtained syrupy mass. Extract this mass with 3 portion of aq. alcohol. Add sufficient quantity of Na_2CO_3 sln to make the extract alkaline. Filter & evaporate Fitrate to obtained dry residue of ephedrine.

(★) 2022(t) How to identify primary & secondary amine in alkaloid?

⇒ Nitrous acid test (HONO):

1. primary amines: React with HONO to produce N_2 gas which can be observed as bubbles.

CamScanner

2. Secondary amines: React with HNO_2 to form Nitrosoamines, which are often yellow in color.

3. Tertiary amines: Do not react with HNO_2.

Hinsberg Test:

1. primary amines: Form a soluble sulfonamide that precipitates upon acidification.

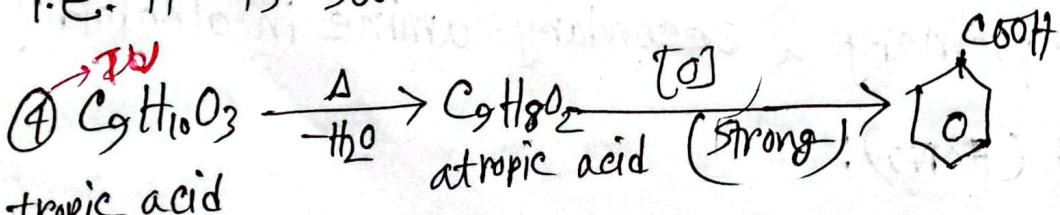
2. ~~Ser~~ secondary amine: Form an insoluble sulfonamide that doesn't dissolve in acid.

3. Tertiary amines: Don't react with benzenesulfonyl chloride.

TU

Discuss structural elucidation of tropic acid?

⇒ (I) Molecular formula is $C_9H_{10}O_3$ (II) DBEV = 5 (III) Tropic acid consumes one equivalent of alkali & does not add bromine i.e. it is saturated monbasic acid.

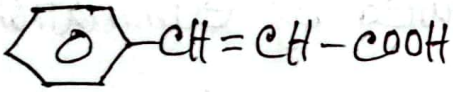


Tropic acid when heated strongly, it lose a molecule of H_2O to yield on unsaturated acid. Further, oxidation gives

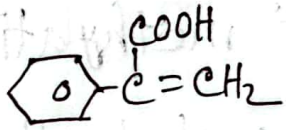
benzoic acid. AS benzoic acid is produce so the molecule has one mono-substituted benzene ring with side chain.

⑤ Tropic acid when acylated forms a monoacylated derivatives, indicating -OH is present.

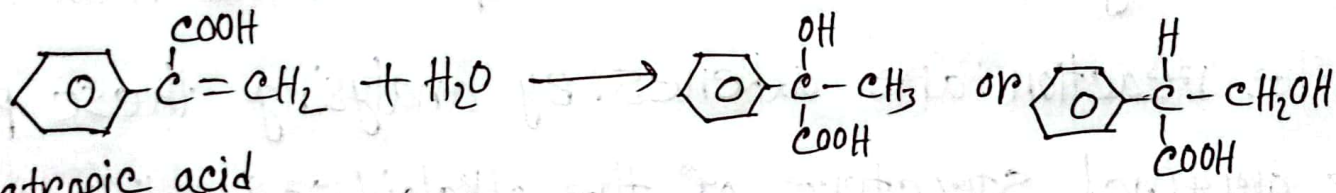
Now, possible structures of Atropic acid are



cinnamic acid



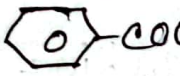
Atropic acid

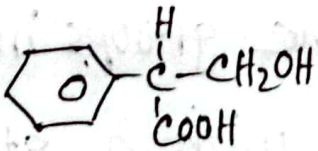
⑥ 

atropic acid

3° Alc (I)

1° Alc (II)

In step-5, By one time oxidation we get  so we can say it has 1° Alc. So the structure II is the real structure of tropic acid. structure:



TU How is heterocyclic system identified in an alkaloid by hofmann ex. method?

⇒ The alkaloid is treated with an excess CH_3I in the

Presence of strong base (KOH), then converting amine groups to quarternary ammonium salt. This salt is heated in the presence of strong base (like Ag_2O) to induce elimination. This step breaks open any Nitrogen containing heterocyclic ring structure present in the molecule. The process of methylation followed by elimination is repeated, ensuring that all 'N' atom are exhausted & any potential cyclic structures are opened. The final products of the reaction are alkenes. By analysing these product the original structure of the alkaloid is identified. so, if a cyclic amine is present this method cleavage of the ring, making it easier to idetify the structure of the heterocyclic system.

TU, 2017

Prove two reactive methylene groups in tropinone?

⇒ when tropinone is treated with a strong base like NaOEt (Sodium ethoxide). The base abstracts a proton from one of these methylene groups, resulting in the formation of an endate ion. The formation of

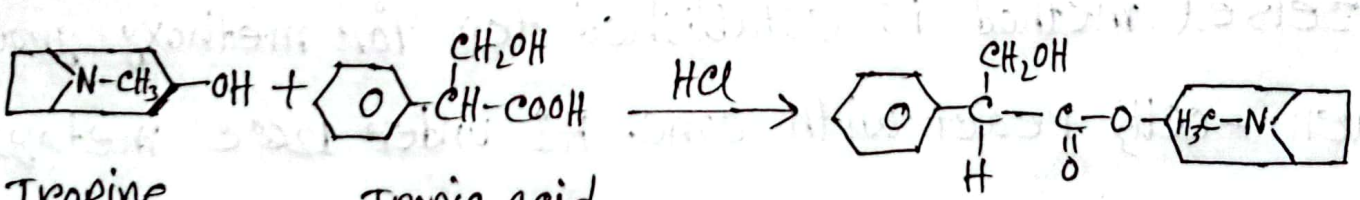
the enolate ion is evidence of the reactivity of the methylene groups.

Tropinone + NaOEt ⟶ Enolate ion

2. Tropinone can undergo a mannich reaction, wher it reacts with formaldehyde & secondary amine like dimethylamine. This results in the formation of a β-amino Ketone, where the methylene group at position 3 or 5 has been substituted. ~~with a~~ This β-amino ketone formation proves that methylene groups are reactive

2016(t)
Show reaction of tropine + tropic acid in presence HCl

⇒



Tropine Tropic acid atropine

2016(t)
prepare morphine from codeine?

⇒ (I) codeine + HBr ⟶ Norcodeine + CH_3Br

(II) Norcodeine + N-bromosuccinimide ⟶ Normorphine + CH_3Br

(III) Normorphine + Me_2SO_4 (NaOH) ⟶ Morphine + $NaHSO_4$

(*) 2016(4) How detect (i) $-OCH_2O-$ (ii) $-OCH_3$?

⇒(i) 1. The methylenedioxy group can be oxidized with $KMnO_4$ in acidic conditions. This reaction cleaves the $-OCH_2O-$ group, yielding HCHO & a product with a carboxylic acid group. The formation of HCHO is detected by its characteristic odor or by using a suitable reagent which forms colored product with HCHO. This HCHO production proves $-OCH_2O-$ group is present.

$$-OCH_2O- + KMnO_4 \text{ (acidic)} \longrightarrow HCHO + \text{carboxylic acid} + MnO_2$$

(ii) zeisel method is established test for methoxy groups. when $-OCH_3$ react with conc. HI under 126°c methyl iodide is formed. The CH_3I is volatile & can be collected in a sln of $AgNO_3$. This forms bright yellow precipitate of AgI. The presence of the yellow precipitate is a strong indicator of a methoxy group.

$$-OCH_3 + HI \xrightarrow{126^\circ} -\overset{+}{\underset{H}{O}}-CH_3 \longrightarrow -OH + CH_3I \xrightarrow{AgNO_3} AgI(\downarrow)$$

2016(H)

Give a synthesis of tropic acid from acetophenone?

$\Rightarrow$ (i) $C_6H_5COCH_3 + Br_2(H^+) \longrightarrow C_6H_5COCH_2Br + HBr$

acetophenone

(ii) $C_6H_5COCH_2Br + KCN \longrightarrow C_6H_5COCH_2CN + KBr$

(iii) $C_6H_5COCH_2CN + H_2O(H^+) \longrightarrow C_6H_5COCH_2COOH + NH_4^+$

Tropic acid